Software Test Plan (STP)

Smart Traffic Complaint and Monitoring System

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Technology baseline: MERN (MongoDB, Express.js, React, Node.js)

Document status: Prepared for submission review; test execution results must be filled only after tests are run.

1. Introduction

1.1 Purpose
This plan defines the scope, approach, environment, responsibilities, entry/exit criteria, and test cases for validating the Smart Traffic Complaint and Monitoring System against its SRS.

1.2 Scope
Testing covers account access, complaint submission and retrieval, administrator workflows, validation, authorization, API errors, basic performance, and browser compatibility for the MERN implementation.

2. Test Strategy
- Functional testing: verify behavior against FR-01 through FR-15.
- Integration testing: verify React UI, Express API, and MongoDB interactions.
- Security testing: verify authentication, ownership checks, role restrictions, input validation, and safe error responses.
- Non-functional testing: execute performance, usability, availability, and browser checks where the required environment is available.
- Regression testing: rerun relevant cases after changes.

3. Test Items and Features to be Tested
- Registration, login, logout.
- Complaint creation and validation.
- Citizen complaint list and status tracking.
- Administrator list, filtering, priority, assignment, status update, dashboard.
- Authentication, role-based authorization, ownership enforcement.
- API error responses, performance, compatibility, and usability.

4. Features Not to be Tested in This Release
- Live integrations with municipal traffic authority systems.
- Automated traffic violation detection or AI classification.
- Real-time traffic sensor feeds.
- Photo upload and live map integrations, unless the team adds them to the core scope.

5. Test Approach, Environment, and Criteria

5.1 Security Validation

ID	Security validation	Expected result
SEC-01	Unauthenticated request to protected complaint endpoint	Request is rejected with 401; no private data returned.
SEC-02	Citizen requests another citizen's complaint	Request is rejected with 403 or a privacy-preserving 404; complaint data is not returned.
SEC-03	Citizen calls administrator endpoint	Request is rejected with 403.
SEC-04	Register/login and inspect stored user record	Password is stored as a salted hash, not plaintext; API does not return hash.
SEC-05	Submit missing or oversized fields	Input is rejected with a controlled 400 response; no record is created.
SEC-06	Send query-operator-like values in input	Input is handled as data and cannot bypass authorization or validation.
SEC-07	Trigger a server-side error in a controlled test environment	Response does not expose stack traces, secrets, or database internals.
SEC-08	Attempt an invalid status update	Server rejects values outside the allowed status list.

5.2 Test Environment

- Application: React frontend, Node.js/Express API, MongoDB database.
- Browsers: current stable Chrome and Firefox.
- Test data: dedicated test users (at least one citizen and one administrator) and seeded complaints.
- Performance test: database seeded with up to 1,000 complaints; record environment and tooling used.
- Do not run destructive or load tests against production data.

5.3 Entry Criteria

- A runnable build is available in a test environment.
- Test accounts and test data are prepared.
- SRS requirements and allowed statuses/priorities are agreed by the team.

5.4 Exit Criteria

- All planned cases have an execution status recorded as Pass, Fail, Blocked, or Not Run.
- No unresolved critical security or data-loss defects remain.
- Failed cases have defect notes and retest status.
- Results are reported honestly; unexecuted cases remain Not Run.

5.5 Pass/Fail Rules

A test passes only when the observed behavior matches the expected result. A test fails when the observed behavior differs. Use Not Run before execution and Blocked when a prerequisite prevents execution.

6. Test Cases

ID	Test	Requirement(s)	Steps / input	Expected result	Execution status
TC-01	Registration	FR-01	Register with valid name, unique email, and valid password.	Account is created; no plaintext password is stored.	Not Run
TC-02	Duplicate registration	FR-01	Register again with an email already in use.	Request is rejected with a clear validation/ conflict message.	Not Run
TC-03	Login/logout	FR-02, SR-03	Log in with valid credentials, then log out.	Login succeeds; logout invalidates the session/token as designed.	Not Run
TC-04	Invalid login	FR-02	Submit incorrect password.	Access is denied; response does not disclose sensitive details.	Not Run
TC-05	Valid complaint submission	FR-03–FR-05	Submit category, description, and location while authenticated.	Complaint is saved with unique ID, timestamp, and Submitted status.	Not Run
TC-06	Missing required field	FR-14	Submit a complaint with location or description missing.	400 validation response; no complaint is saved.	Not Run
TC-07	Citizen complaint list	FR-06, FR-07	Log in as a citizen and open the complaint list.	Only that citizen's complaints and statuses are shown.	Not Run
TC-08	Cross-user access	SR-01, NFR-09	Attempt to retrieve a different citizen's complaint ID.	Private complaint data is not returned.	Not Run
TC-09	Admin-only endpoint	FR-15, SR-02	Call admin complaint endpoint as a citizen.	403 Forbidden; no admin data is returned.	Not Run
TC-10	Filtering	FR-08, FR-09	Filter administrator complaint list by status and category.	Only records matching the selected filters are returned.	Not Run
TC-11	Priority and assignment	FR-10, FR-11	Admin sets allowed priority and assigns a complaint.	Updated values are saved and shown.	Not Run
TC-12	Status update	FR-12	Admin updates status to an allowed value.	Status changes and is visible in the complaint detail/list.	Not Run
TC-13	Invalid status	FR-12, SR-04	Submit a status value not in the allowed list.	Request is rejected; existing status remains unchanged.	Not Run

TC-14	Dashboard counts	FR-13	Compare dashboard totals with known seeded complaint records.	Counts match the test dataset.	Not Run
TC-15	Performance	NFR-01	With up to 1,000 complaints, measure 100 standard list/detail requests.	At least 95% complete within 3 seconds in the recorded test environment.	Not Run
TC-16	Browser compatibility	NFR-04	Run core flows in current stable Chrome and Firefox.	Core flows operate without blocking browser-specific defects.	Not Run
TC-17	Usability	NFR-02	Ask five first-time users to submit a valid complaint using instructions.	At least four finish within 3 minutes after login.	Not Run
TC-18	Controlled errors	NFR-06, SR-06	Cause a controlled validation or test-environment server error.	No false success; safe, documented error response is returned.	Not Run

7. Requirement Traceability

SRS requirement(s)	**Verification test(s)**
FR-01–02	TC-01–04
FR-03–05	TC-05–06
FR-06–07	TC-07–08
FR-08–09	TC-10
FR-10–12	TC-11–13
FR-13	TC-14
FR-14	TC-06, TC-18
FR-15	TC-09
NFR-01	TC-15
NFR-02	TC-17
NFR-03	Availability acceptance run; Not Run until executed
NFR-04	TC-16
NFR-06	TC-18
NFR-07	TC-05, TC-14
NFR-08	TC-01, TC-03, SEC-04
NFR-09	TC-08, TC-09, SEC-01–SEC-03
NFR-10	TC-06, TC-13, SEC-05, SEC-08

8. Roles and Deliverables

- Test lead: coordinate execution and consolidate results.
- Developers: fix defects and provide test builds.
- All team members: review requirements, execute assigned cases, and record evidence.
- Deliverables: this plan, test execution record, defect list, and final test summary.